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Subject: Programming with Java	
ASSIGNMENT No : 2	
TITLE : Constructors in Java	

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

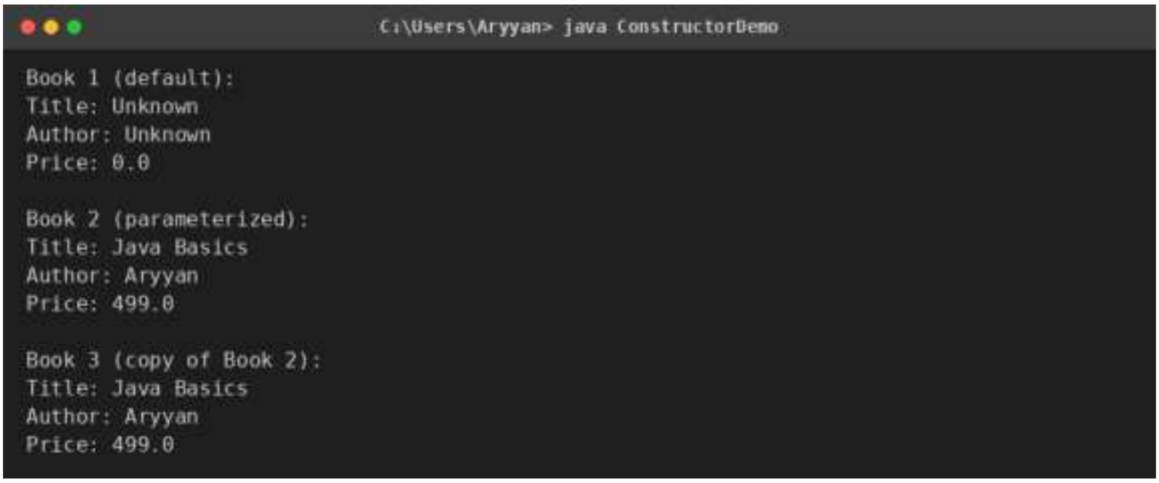
To understand and implement default, parameterized, and copy constructors for object initialization.

Programme

```
class Book {
    String title;
    String author;
    double price;

    // default constructor
    Book() {
        title = "Unknown";
        author = "Unknown";
        price = 0.0;
    }

    // parameterized constructor
    Book(String title, String author, double price) {
        this.title = title;
        this.author = author;
        this.price = price;
    }

    // copy constructor
    Book(Book b) {
        this.title = b.title;
        this.author = b.author;
        this.price = b.price;
    }
```

```
    void display() {
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Price: " + price);
    }
}

public class ConstructorDemo {
    public static void main(String[] args) {
        Book b1 = new Book();                                  // default
        Book b2 = new Book("Java Basics", "Aryyan", 499.0);     // parameterized
        Book b3 = new Book(b2);                                 // copy

        System.out.println("Book 1 (default):");
        b1.display();

        System.out.println("\nBook 2 (parameterized):");
        b2.display();

        System.out.println("\nBook 3 (copy of Book 2):");
        b3.display();
    }
}
```

Output



Conclusion

Thus, a Java program was developed and executed to demonstrate default, parameterized, and copy constructors. It was observed that the default constructor initializes an object with pre-set values, the parameterized constructor allows values to be supplied at the time of object creation, and the copy constructor creates a new object by duplicating an existing object's data. This helped in understanding how constructor overloading enables flexible and appropriate object initialization depending on the situation.